

OrganChip Insight

A traceable microscopy workspace for organ-on-chip experiments

TOOL & PLATFORM | AI4S OPEN INNOVATION 2026

Evidence-gated image analysis that keeps sources, overlays, measurements, provenance, uncertainty, and negative findings in one reproducible local experiment.

Candidate date	17 September 2026
Team	Solo team; public author name pending
Release	Validated on dev; public tag pending authorization
Scientific scope	Non-clinical, exploratory microscopy workflow

Candidate PDF. Apache-2.0 is fixed for the code; the public author name, release tag, and public URLs remain owner-controlled fields.

Abstract

Organ-on-chip microscopy produces valuable longitudinal images, but an image analysis result is difficult to trust or reuse when acquisition metadata, algorithm identity, computational cost, and scientific limits are separated from the output. OrganChip Insight is a local, reproducible Tool & Platform that combines experiment management, robust image import, CPU segmentation, overlay inspection, per-image measurements, result export, and a transparent engine registry. The default adaptive engine requires no learned weights and operates offline. External BBBC019 and BBBC038 benchmarks are generated by versioned scripts and surfaced in the product with uncertainty, runtime, memory, promotion decisions, and SHA-256 provenance.

Credibility is part of the implementation rather than an appendix. An organ-on-chip quality-classification study used grouped splits, locked runtime contracts, metadata-only comparators, mode-specific metrics, and preregistered ablations. No configuration demonstrated a robust quality signal independent of acquisition and culture metadata, so the frozen test split was never opened. The selected CNN is limited to an optional ONNX demonstrator that always abstains and never produces an automatic good/bad decision. Similarly, the interface reports connected components rather than cells because no instance annotation validates counting on the competition images.

The resulting system is not a diagnostic device. It is an auditable local workflow for quality-aware microscopy exploration, designed to preserve the relationship between images, measurements, metadata, uncertainty, and negative findings.

1. Problem and target user

Bright-field organ-on-chip experiments often produce repeated images across devices, lineages, culture days, and acquisition modes. A laboratory user needs more than an isolated segmentation model. The useful unit is an experiment that records context, validates incoming files, preserves raw images, connects every output to an engine version, and exports evidence that can be reviewed outside the application.

The target user is a research laboratory working with non-clinical organ-on-chip microscopy. The primary workflow is:

- 1 create a named experiment and record available acquisition metadata;
- 2 import an image batch and inspect accepted and rejected files;
- 3 run a declared analysis engine;
- 4 review overlays and per-image measurements;
- 5 export structured evidence for downstream analysis or audit.

OrganChip Insight does not claim diagnosis, treatment recommendation, toxicity prediction, efficacy prediction, or validated cellular counting.

2. Data, provenance, and legal basis

2.1 Datasets

Dataset	Version and licence	Project use	Traceability
OOC Image Dataset, Zenodo 10203721	2023 release, CC-BY-4.0	grouped quality-classification study	locked train/validation/test manifests and checksums
BBBC019 Microfluidics	v2, CC-BY-3.0	external foreground-segmentation benchmark	dataset manifest, per-image CSV, JSON report
BBBC038 stage 1 train	v1, CC0-1.0	bounded instance and count benchmark on 12 images	archive, image and mask-tree hashes; locked subset manifest

The full data declaration is maintained in [docs/declaration-ia-et-licences.md](#). No personal, clinical, or private data is used.

2.2 Why three datasets are not one validation claim

The datasets answer different questions. BBBC019 provides binary foreground masks for a small microfluidic DIC set. BBBC038 provides nuclear instance masks and can audit instance detection and counting. The competition OOC dataset provides image-level quality labels but no instance masks. Scores are therefore presented within each dataset and task. They are never merged into a single performance number, and BBBC019 or BBBC038 does not validate cell counting on the OOC images.

2.3 Longitudinal dependence and split design

Filename and acquisition structure showed that adjacent dates and shared lineages could cross a naive split. The campaign audit identified 29 three-day campaigns and 270 test images exposed to a neighboring date with their own lineage under the earlier split. This is a risk of longitudinal dependence, not proof that two images come from the same physical chip. A campaign-aware v2 manifest was therefore preregistered and generated while the v1 split was preserved. Details are in [docs/protocole-campagnes-v2.md](#) and [reports/ooc-campaign-split-v2.json](#).

The honest statement remains: the split is grouped by acquisition date and controlled for detected near-duplicates, but biological independence is not established because chip and well identifiers are absent.

3. System architecture

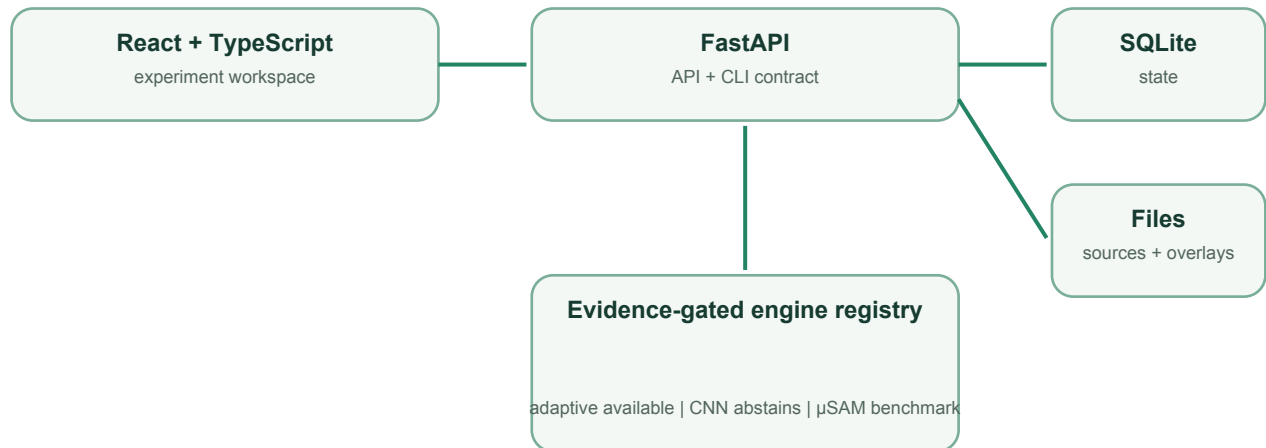


Figure 1. Local architecture and evidence-gated engine roles.

The platform uses a deliberately small local architecture:

```

text
React + TypeScript interface
|
v
FastAPI application ---- OpenAPI and CLI
|
+---- SQLite: experiments, images, results
+---- filesystem: uploads and overlays
+---- engine registry
      +---- adaptive segmentation v1
      +---- optional run B ONNX demonstrator
      +---- isolated benchmark candidates
  
```

Docker Compose packages the frontend, Nginx, backend, and persistent data volume. Nginx and the API enforce aligned size and security constraints. The containers run as non-root users. Exact Python and JavaScript dependencies are locked in **backend/uv.lock** and **frontend/package-lock.json**.

4. Product method

4.1 Robust import

The backend decodes PNG, JPEG, and TIFF rather than trusting file extensions. It bounds file size and total pixel count, detects binary duplicates, and supports grayscale 16-bit TIFF. Measurement input is normalized by robust percentiles while the original file remains preserved. Import and analysis are separate operations, which prevents a failed analysis from duplicating images on retry.

4.2 Adaptive segmentation

The default engine uses classical thresholding and morphology, followed by connected-component analysis. It has no external weights and runs on CPU. Outputs include the segmentation overlay, foreground fraction, component count, area summaries, estimated diameter in pixels, intensity, contrast, and a relative contrast index.

These outputs are exploratory heuristics. In particular, a connected component may combine multiple cells or represent debris, background texture, or another structure. The interface and exports retain the term “connected component.”

4.3 Engine registry and experimental CNN

Each engine declares its status, availability, intended role, and limitations. The run B MobileNetV3 model can only be enabled from a local ONNX bundle whose model, preprocessing, and selection-report hashes match the declared values. It resizes a grayscale input to 448 pixels according to the exported preprocessing contract.

Every CNN output has status “Needs review.” The softmax is explicitly labelled uncalibrated, acquisition mode and provenance hashes are shown, and no automatic good/bad conclusion, segmentation overlay, or count is attached. Without ONNX Runtime or the exact bundle, the engine remains visible but unavailable and the adaptive workflow continues to work.

5. Evaluation design

5.1 External segmentation and instance benchmarks

The BBBC019 scripts evaluate foreground F1 and IoU per image, bootstrap macro metrics, record inference time and process memory, and generate error overlays. Adaptive and μ SAM reports cover the same 13 images.

The BBBC038 protocol was committed before execution. It fixes a deterministic 12-image subset, μ SAM `vit_b_lm` with automatic prompt generation, object F1 at IoU 0.50 and 0.75, median absolute percentage count error, and three promotion criteria. No post-result tuning was performed. See [docs/protocole-bbbc038-instances-v1.md](#).

5.2 Quality classification

The classification target is binary **good/bad**. Culture-day bucket and lineage are covariates and comparators, not prediction targets. Validation was reported globally and by acquisition mode, resolution, lineage, and day.

The workflow separated smoke tests, validation, and final evaluation. Runtime, source bundle, ImageNet weights, split manifests, checkpoints, and result archives were identified by hashes. A reference color model at 224 pixels and two preregistered ablations were evaluated: grayscale at 224 pixels (A) and grayscale at 448 pixels (B). A third crop ablation was conditional and was not run because the preregistered RGB criterion was not reached. The frozen test set was never opened.

Metadata-only rules were fit on the training split and evaluated on validation to measure shortcut strength. This matters because a globally strong CNN can still fail within an acquisition mode.

6. Results

6.1 Foreground segmentation on BBBC019

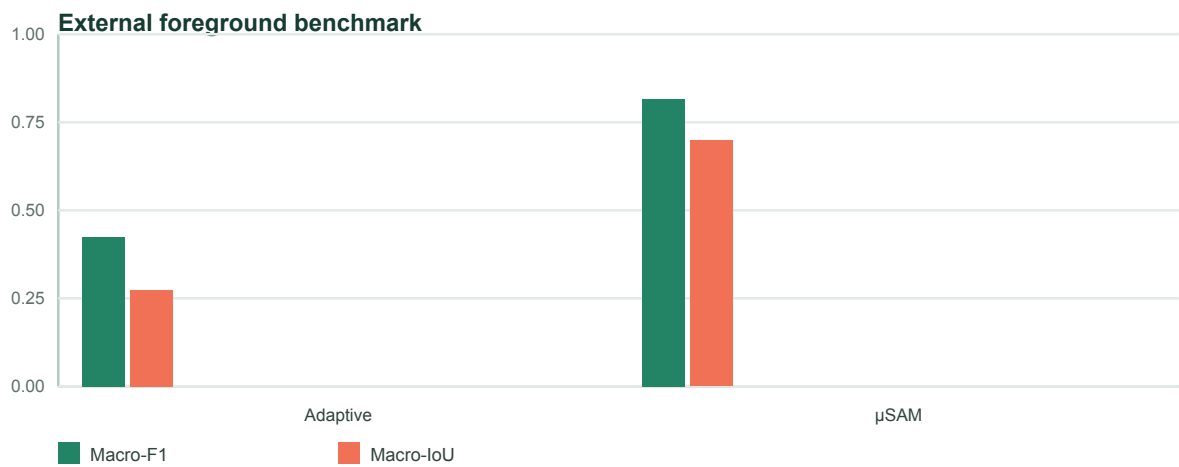


Figure 2. BBBC019 macro scores; the stronger μSAM result carries much higher measured cost.

Engine	Macro-F1	95% bootstrap interval	Macro-IoU	Mean CPU time/image	Peak process memory
Adaptive segmentation v1	0.424892	[0.369769, 0.467653]	0.273632	0.112333 s	324.668 MB
μSAM vit_b_lm APG	0.815542	[0.758206, 0.861862]	0.698535	42.843791 s	8702.957 MB

Source reports: [adaptive JSON](#) and [μSAM JSON](#).

μSAM is more accurate on this small external foreground task, but is about 381 times slower per image in the measured environments and uses substantially more memory. Runtime ratios must be read as environment-specific, not a general hardware-independent speed claim.

6.2 Instance audit on BBBC038

Metric	Result
Macro object F1 at IoU 0.50	0.628327
Macro object F1 at IoU 0.75	0.481698
Median absolute percentage count error	15.3409%
Mean CPU time/image	36.690719 s
Peak process memory	8896.457 MB

The source report is [bbbc038-stage1-subset-v1-microsam-vit-b-lm-apg.json](#). Two of the three preregistered promotion criteria failed. μ SAM was not promoted to the product, and no tuning was performed after observing the result.

6.3 Quality classification and metadata shortcuts



Figure 3. Mode-specific validation exposes the weak and unstable RGB signal hidden by global scores.

Run	Global BA	L BA	RGB BA	Global AUC	L AUC	RGB AUC
Reference color 224	0.7619	0.7729	0.6315	0.7996	0.8521	0.6309
A: grayscale 224	0.6704	0.7389	0.5948	0.7176	0.7529	0.5848
B: grayscale 448	0.7750	0.8338	0.6079	0.8414	0.8551	0.6985
Metadata: mode × culture-day bucket	0.7920	0.8171	0.6533	-	0.8171	0.7068

The metadata comparator values require an important sensitivity statement. In RGB, its 0.6533 balanced accuracy comes entirely from the distinction between the published labels **4_days** and **4+_days**; merging those labels reduces RGB balanced accuracy to 0.5000 and global balanced accuracy to 0.7413. The comparator is evidence of dataset structure and campaign identification, not a deployable quality model.

For run B, the bootstrap difference from the reference in RGB includes zero, and leave-one-date-out threshold selection is unstable. Post-hoc within-stratum analysis finds only a modest in-sample RGB residual and cannot separate it from model-selection effects. Full evidence and corrections are in the [classification counter-audit](#) and the [versioned comparator report](#).

Fixed scientific conclusion: “Sur cette validation, aucun signal de qualité robuste et indépendant des métadonnées d'acquisition et de culture n'est démontré ; le résidu observé en RGB pour le run B est modeste, mesuré en échantillon et non distinguable d'un effet de sélection.”

Therefore the classification study is closed, no configuration is eligible, and the frozen test set remains unopened.

7. Product verification

The backend suite contains 177 collected tests in the 17 September 2026 release candidate: 176 pass and one environment-dependent test is skipped. Ruff, TypeScript type-checking, the production Vite build, notebook synchronization, benchmark summary synchronization, release-tree audit, checksum inventory, and Docker Compose configuration pass through **make check**.

Playwright starts isolated Docker projects and fresh volumes. The fast path creates an experiment, imports a synthetic image, runs adaptive inference, checks the overlay and scientific reservation, and downloads JSON and CSV. A second path imports three hash-locked public images (OoC RGB, OoC grayscale, and BBBC019 TIFF), verifies source-format metadata, runs the same workflow, and confirms that all source hashes remain unchanged. Additional gates report zero Axe violations, verify visible keyboard focus, mobile reflow at 390 px, an effective 200% layout width, and persistence of sources, results, exports, gallery, and viewer after both containers restart.

The candidate was also rebuilt with **docker compose build --pull --no-cache**. The release audit found no tracked secret, personal path, cache, raw dataset, or undistributable model binary. The SHA-256 inventory covers 40 tracked reports, manifests, notebooks, submission sources, and frontend evidence.

8. Credibility and limitations

- 1 **Biological grouping.** Acquisition date is a proxy, not a chip or well identifier. Biological independence cannot be established from the released metadata.
- 2 **Quality labels.** Adjacent or near-duplicate-looking images can have conflicting labels. The project does not reinterpret those labels after observing model outputs.
- 3 **Acquisition shortcuts.** Resolution nearly identifies acquisition mode, and mode interacts strongly with class prevalence. Global metrics alone are insufficient.
- 4 **Small external benchmarks.** BBBC019 contains only 13 evaluated images; BBBC038 uses a preregistered 12-image audit subset. Intervals are shown, and neither result is a population claim over OOC images.
- 5 **No physical calibration.** Areas and diameters are in pixels, not μm or μm^2 .
- 6 **No validated cell count.** Connected components are heuristic image structures, not verified cells or nuclei on the OOC dataset.
- 7 **Uncalibrated CNN output.** Softmax values from the optional demonstrator are not calibrated probabilities and never drive an automatic decision.
- 8 **Deployment scope.** SQLite and local files are appropriate for a local research prototype or a single instance, not a multi-tenant clinical system.

9. Practical value

The platform's immediate value is workflow discipline: images, metadata, engine identity, overlays, measurements, exports, and limitations live in one experiment. A laboratory can inspect why an image was rejected, rerun an analysis without reimporting files, compare engine evidence, and export a reviewable record without an external service.

The negative CNN result is also operationally useful. It prevents a shortcut- driven classifier from becoming an automatic gate and identifies the metadata needed for future datasets: stable culture-day vocabulary, chip and well IDs, explicit artifact types, physical calibration, and at least five independent dates per acquisition mode.

Future research may study a maturation-deviation score conditioned on mode, day and lineage; adversarial auxiliary objectives for culture day; expert artifact annotations; and OME-Zarr interoperability. These are perspectives, not implemented submission features.

10. Reproduction

10.1 Product

```
bash
cp .env.example .env
docker compose up --build
```

Open <http://localhost:8080>. The API is at <http://localhost:8000> and the OpenAPI document at <http://localhost:8000/docs>.

10.2 Verification

```
bash
make check
npm --prefix frontend exec playwright install chromium
make test-e2e
make test-real-images
make test-e2e-real
make test-release-persistence
```

10.3 Command-line inference

```
bash
cd backend
uv run python inference.py image-1.png image-2.tif --output-dir artifacts/demo
```

10.4 Versioned benchmark view

```
bash
make benchmark-summary
```

make check verifies that the generated frontend summary exactly matches the three locked JSON reports. Raw datasets remain outside Git and are acquired or verified through the versioned manifests and scripts documented in [data/README.md](#).

11. AI tools, libraries, and licences

OpenAI Codex and Anthropic Claude Code supported code, tests, documentation, independent review, and literature review under human control. Each reported metric comes from a versioned script or report; generated prose is not presented as experimental evidence. The product does not depend on an LLM, hosted model, or commercial API.

The canonical disclosure, including μ SAM, MobileNetV3, datasets, and main libraries, is [docs/declaration-ia-et-licences.md](#).

Code licence: Apache-2.0, recorded in the repository-root **LICENSE** file.

Model bundle: not distributed in this submission. Any later publication requires a separate licence decision and hash-locked bundle.

12. Conclusion

OrganChip Insight demonstrates that a useful organ-on-chip imaging platform is not defined by a single model score. It is defined by a complete path from validated input to inspectable and exportable evidence, with provenance and limitations preserved. The adaptive workflow is fast, local, and available; stronger but costly external segmentation remains a benchmark; and a shortcut-sensitive classifier is prevented from making automated decisions. The platform is therefore technically complete enough to demonstrate, while remaining explicit about the scientific validation that future data must provide.

References

- 1 Movčana et al. "Organ-On-A-Chip (OOC) Image Dataset for Machine Learning and Tissue Model Evaluation." *Data* 9(2):28, 2024.
- 2 Ljosa, Sokolnicki, and Carpenter. "Annotated high-throughput microscopy image sets for validation." *Nature Methods* 9:637, 2012.
- 3 Archit et al. "Segment Anything for Microscopy." *Nature Methods*, 2025.
- 4 Archit and Pape. "Revisiting foundation models for cell instance segmentation." MIDL, 2026.
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